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Research Topic: Edge-AI Multimodal Fetal Ultrasound for Maternal Risk Stratification in Resource-Constrained Rural Bangladesh

Title: Edge-AI Multimodal Fetal Ultrasound for Maternal Risk Stratification in Resource-Constrained Rural Bangladesh

1 Research Problem

1.1 Socio-Infrastructure Bottlenecks: Despite the improvements in the maternal and perinatal mortality rates, the issue continues to be a public health problem in rural Bangladesh. One of the main reasons for this crisis is the grossly under-distributed trained obstetricians and specialist sonographers in the Union Health and Family Welfare Centres [1]. The socio-cultural barriers and the high cost of travelling to the urban tertiary care centres are major issues in reaching pregnant women in remote villages in time. Moreover, the power grid at rural health care centers is not stable and there is frequent load shedding. This makes it impractical and also limits Community Health Workers who are not specialists to have the diagnostic tools required to detect high-risk pregnancies before complications ensue with conventional high-voltage ultrasound equipment.

1.2 Demographic Data Bias: Built-in bias in demographic data is compounded by the use of existing artificial intelligence (AI) diagnostic models, continuing to worsen health challenges. Most of the commercial machine learning models are trained on fetal biometric data from Europe and North America. The literature shows the natural, lower baseline measurements of parameters like femur length and abdominal circumference of healthy south Asians fetuses and the lower maternal BMI [2]. This statistical mismatch of distribution of patients leads to a systematic overestimation of risk when western trained algorithms are used in evaluating Bengali patients. In practice, when the mean biometric value μ_{global} of the data in question is much lower than the mean value μ_{local} of the training data, the predictive model is bound to make predictions with high false positive rates for the condition, such as Intrauterine Growth Restriction. This structural bias creates unwarranted panics and inefficient use of precious tertiary medical care resources.

1.3 Operational Need: To correct for this failure of the system in diagnostics, a paradigm shift in point-of-care medical technology is needed. An operational requirement for a low-power diagnostic framework for use in resource constrained environments is in urgent need. Community Health Workers need a screening tool which is offline first, meaning it does not need constant Internet access or consistent grid power. A targeted mobile diagnostic intervention that can accurately stratify maternal risk directly in rural communities by overcoming the limitations of models globalized for AI applications.

2 Existing Literature

2.1 Global Fetal Growth Standards and South Asian Biometrics: In recent epidemiological studies, globalized standards, like fetal growth standards set by INTERGROWTH-21st [3] are largely used. Although these databases offer a common set of variables for the whole world, recent studies at the regional level have shown some variations for South Asian population. Research shows that when generally accepted reference values for biometric thresholds are used, the incidence of fetal growth restriction is increased in the Bengali and South Asian population [4]. There are no currently available strong predictive models of physiological baseline to which the natural systems of these localized populations can be calibrated.

2.2 Point-of-Care Ultrasound and Edge AI: It has made great progress to introduce CNN (convolutional neural networks) to portable ultrasound devices [5]. Lightweight networks such as MobileNetV3 has been shown to be very effective in real-time medical image classification [6]. The vast majority of these systems, however, rely on arrays of cloud computing systems to achieve acceptable speeds of inference. This is because they rely on constant internet connectivity to be useful, which is not always possible in clinical settings in remote areas. Furthermore, existing AI devices for diagnosis are limited to analyzing visual data from a single modality and are not yet able to consider the relevant physiological data that obstetricians routinely consider when interpreting ultrasound images, namely, maternal information.

2.3 The Literature Gap: The survey of existing technologies shows a clear gap in existing diagnostic technology for resource-limited environments.There is no existing methdology that offers an architecture that processes live ultrasound B-scans and maternal vital signs on an offline-first basis multimodal Edge-AI framework. Moreover, there is no standard protocol which is available to calibrate such fused neural network especially for South Asian fetal biometrics and which works in limited computational resources such as the low cost mobile processors [7]. This study is filling in that very niche of localized clinical data and optimized edge computing.

3 Research Question

3.1 Primary Question: Can an offline-first multimodal Edge-AI framework on low cost mobile hardware be used to improve the accuracy of maternal risk stratification by non-specialist health workers in rural Bangladesh, and reduce false positive fetal growth restriction diagnoses?

3.2 Sub-Questions and Hypotheses: The overall question is complemented by three specific sub-questions around biometric calibration, multimodal data fusion and edge computing optimization.

Sub-Question 1: What are the quantitative clinical benefits of calibration of the diagnostic algorithms to local (rural Bengali) fetal biometrics versus global reference data sets in terms of reduction in false-positive diagnosis? The hypothesis is that localized calibration will make a substantial difference in the diagnostic threshold mismatch that has been seen in different cohorts of South Asian countries.

Sub-Question 2: Is there statistically significant improvement of the risk stratification from the feature fusion of live ultrasound images and maternal vital signs over image-only models? The goal is to maximise a diagnostic performance measure mathematically expressed as:

$$AUC = \int_0^1 TPR \; d(FPR)$$

Where TPR is the true positive rate and FPR is the false positive rate. The hypothesis is that if maternal vitals are added, then there will be a greater Area Under the Curve.

Sub-Question 3: What are the consequences on inference latency and thermal management of using 8-bit integer quantization after training on small budgets Android processors? The technical hypothesis assumes that the use of the quantization strategy will not affect the clinical diagnostic sensitivity and will introduce a minimal

hardware delay, and therefore enable real-time video processing with a latency of below 100 milliseconds.

4 Methodology

4.1 Localized Dataset curation and annotation: The first step of the process is to collect a large training set of at least $N \geq 5000$ second and third trimester ultrasound B-scans. The Ultrasound videos would be collected from Tertiary referral hospitals. Senior radiologists will manually mark all the imagery to create a robust ground truth and link the specific biometric indicators with documented maternal clinical outcomes, against South Asian biometric baselines.

4.2 Multimodal Architecture & Quantization: A dual-branch neural network design is used in the system. The vision processing branch uses a relatively efficient MobileNetV3 architecture [6] to extract spatial features from the live ultrasound frames. At the same time, Multi-Layer Perceptron processes the structured vital signs of the mother e.g., blood pressure and gestational age. These different data flows are combined at a late stage fusion layer to provide a complete risk probability. The final model is quantized to 8-bit integers using TensorFlow Lite for post training[7]. This INT8 conversion allows to achieve reduced memory footprint and avoid thermal throttling on low cost Android devices without compromising on the diagnostic accuracy.

4.3 System Workflow

All the processes in this architecture go through four consecutive steps as shown in Figure 1 below. Initially, live B-scan frames and maternal parameters go to the dual branches. Secondly, the fusion layer makes this all task together to come up with a single risk score that's processed by the quantized engine. Third, the uncertainty module evaluates the predictive confidence C against the mathematical threshold. Fourth, depending on that score, the device either takes a minimal text record locally for high-confidence results, or encrypts and caches the compressed video for deferred Wi-Fi or mobile data transmission for complex edge-cases.

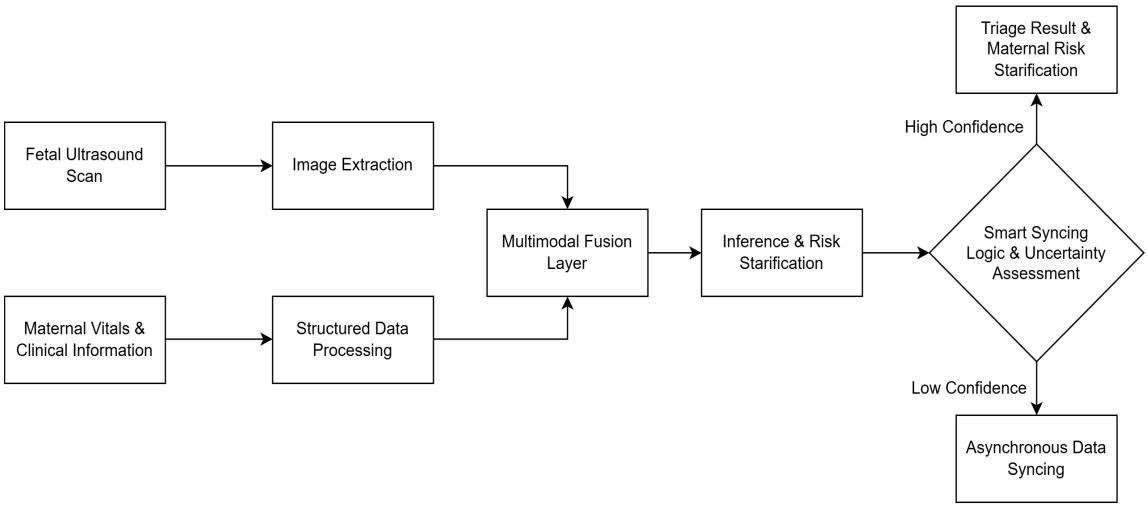


Figure 1. Proposed operational workflow for the Edge-AI multimodal fetal ultrasound system, showing branches feature extraction, quantization, decision thresholding and smart syncing protocol.

4.4 Uncertainty-Driven Smart Syncing Protocol: The application's behaviour for local storage and data transmission is controlled by a mathematical confidence threshold. Each diagnostic event has a predictive confidence score C calculated. If the output confidence is $C \geq 0.85$, the app removes the high bandwidth video stream locally, thereby keeping only a very small amount of text log of the encounter. On the other hand, if the value of $C < 0.85$, the application keeps the ultrasound frames and patient data in the cache. When stable Wi-Fi or internet is available, these complex edge cases will be uploaded asynchronously by the device.

4.5 Field Validation & Usability Tests: The validation process includes a pilot implementation of the intervention in five Upazila Health Complexes with 30 Community Health Workers. The main performance criterion is the sensitivity of the diagnosis made by non-specialist operators compared with the diagnosis made by senior obstetricians baseline. In addition, the field test assesses the usability of the software application in the field when hardware loads are simulated and when there is a local power outage.

4.6 Ethical Considerations: Before conducting any research, all research methods must be formally approved by a recognized Institutional Review Board. All participants are required to give informed, written consent for the study. The edge software, to secure patient privacy, encrypts all data files that are stored in local cache using AES-256 and ensures that the data is anonymized before any data is sent asynchronously to the cloud.

5 Research Topic

5.1 Regional Urgency and policy Alignment: The study can be considered a relevant public health issue in Bangladesh. The project also aligns with the national health policies of decreasing the maternal and neonatal death rates in rural unions [1]. This intervention enables the capacity to diagnose conditions such as preeclampsia and Fetal Growth Restriction to be delivered to the field, thereby closing the gap between the diagnosis and treatment of such conditions in rural primary care and tertiary health in urban areas [2].

5.2 Scalability for Resource Constrain Settings: This project will not only serve as a blueprint for the deployment of the same in Bangladesh but will be a scalable blueprint for the world of health deployment. Combining zero-connectivity Edge-AI, INT8 post-training quantization [7] and an uncertainty-driven smart-syncing protocol enables a system design that can be applied to other low and middle earn countries that are affected by similar challenges, such as lack of specialists and unreliable power supplies.

6 Expected Outcomes

6.1 Diagnostic Accuracy and Calibrated Triage: The major clinical endpoint is the decrease in false-positive intrauterine growth restriction IUGR diagnosis in the rural Bangladeshi populations. Substituting the global algorithms with localized biometric calibration models will be helpful to increase the specificity of the diagnosis and the use of ultrasound feature extraction along with the vital sign parameters of the mother is expected to give a statistically significant improvement in the overall prediction

Area Under the Curve.

6.2 Technical Feasibility on Edge Hardware: The 8-bit integer quantized multimodal model will run entirely offline on low cost Android devices with real-time inference latency below 100ms without thermal throttling or need for continuous electricity.

6.3 Community Healthcare Capacity Building: The Community Health Workers of the Union Health and Family Welfare Centres in the rural areas who are not proficient in using laboratory equipment, will receive an easy-to-use mobile point-of-care decision tool to allow for accurate early maternal risk stratification and to close the diagnostic divide between the remote villages and urban tertiary hospitals.

Quality of Writing

The writing of this proposal clear and appropriate for the multidisciplinary judging panel. Technical jargon is avoided and kept to a minimum when used, and it is carefully contextualized in order to avoid confusion. The proposal has been written in accordance with the formal research conventions, followed the English grammatical rules and followed IEEE citations rule and meets all the IARCO submission requirements.

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